

CSB24084(Assignment 12)

```
from abc import ABC, abstractmethod
import json
import os
```

```
class PaymentMethod(ABC):
    @abstractmethod
    def pay(self, amount: float):
        pass
```

```
class CreditCardPayment(PaymentMethod):
    def pay(self, amount: float):
        print(f"Paid ₹{amount} using Credit Card")
```

```
class UPIPayment(PaymentMethod):
    def pay(self, amount: float):
        print(f"Paid ₹{amount} using UPI")
```

```
class WalletPayment(PaymentMethod):
    def pay(self, amount: float):
        print(f"Paid ₹{amount} using Wallet")
```

```
class NotificationService(ABC):
    @abstractmethod
    def send(self, message: str):
        pass
```

```
class EmailNotification(NotificationService):
    def send(self, message: str):
        print(f"Email sent: {message}")
```

```
class SMSNotification(NotificationService):
    def send(self, message: str):
        print(f"SMS sent: {message}")
```

```
class PushNotification(NotificationService):
    def send(self, message: str):
```

```
print(f"Push notification: {message}")
```

```
class OrderRepository(ABC):  
    @abstractmethod  
    def save(self, order):  
        pass
```

```
class JSONFileRepository(OrderRepository):  
    def __init__(self, filename="orders.json"):  
        self.filename = filename  
  
    def save(self, order):  
        data = []  
  
        if os.path.exists(self.filename):  
            with open(self.filename, "r") as f:  
                try:  
                    data = json.load(f)  
                except:  
                    data = []  
  
        order_data = {  
            "order_id": order.order_id,  
            "amount": order.amount,  
            "final_amount": order.get_final_amount(),  
            "type": order.__class__.__name__  
        }  
  
        data.append(order_data)  
  
        with open(self.filename, "w") as f:  
            json.dump(data, f, indent=4)  
  
        print(f"Order {order.order_id} saved to JSON file")
```

```
class Order(ABC):  
    def __init__(self, order_id: int, amount: float):  
        self.order_id = order_id  
        self.amount = amount  
  
    @abstractmethod
```

```
def get_final_amount(self) -> float:
    pass
```

```
class RegularOrder(Order):
    def get_final_amount(self) -> float:
        return self.amount
```

```
class DiscountedOrder(Order):
    def __init__(self, order_id: int, amount: float, discount: float):
        super().__init__(order_id, amount)
        self.discount = discount

    def get_final_amount(self) -> float:
        return self.amount - self.discount
```

```
class PriorityOrder(Order):
    def get_final_amount(self) -> float:
        return self.amount + 50
```

```
class OrderService:
    def __init__(self, payment: PaymentMethod,
                 notifier: NotificationService,
                 repository: OrderRepository):
        self.payment = payment
        self.notifier = notifier
        self.repository = repository

    def place_order(self, order: Order):
        final_amount = order.get_final_amount()
        self.payment.pay(final_amount)
        self.repository.save(order)
        self.notifier.send(
            f"Order {order.order_id} placed successfully! Amount: ₹{final_amount}"
        )
```

```
if __name__ == "__main__":
```

```
    order_id = int(input("Enter Order ID: "))
    amount = float(input("Enter Amount: "))
```

```
print("\nSelect Order Type:")
print("1. Regular")
print("2. Discounted")
print("3. Priority")
order_type = int(input("Choice: "))

if order_type == 2:
    discount = float(input("Enter Discount: "))
    order = DiscountedOrder(order_id, amount, discount)
elif order_type == 3:
    order = PriorityOrder(order_id, amount)
else:
    order = RegularOrder(order_id, amount)

print("\nSelect Payment Method:")
print("1. Credit Card")
print("2. UPI")
print("3. Wallet")
p = int(input("Choice: "))

if p == 1:
    payment = CreditCardPayment()
elif p == 2:
    payment = UPIPayment()
else:
    payment = WalletPayment()

print("\nSelect Notification Type:")
print("1. Email")
print("2. SMS")
print("3. Push")
n = int(input("Choice: "))

if n == 1:
    notifier = EmailNotification()
elif n == 2:
    notifier = SMSNotification()
else:
    notifier = PushNotification()

repository = JSONFileRepository()

service = OrderService(payment, notifier, repository)
```

service.place_order(order)